

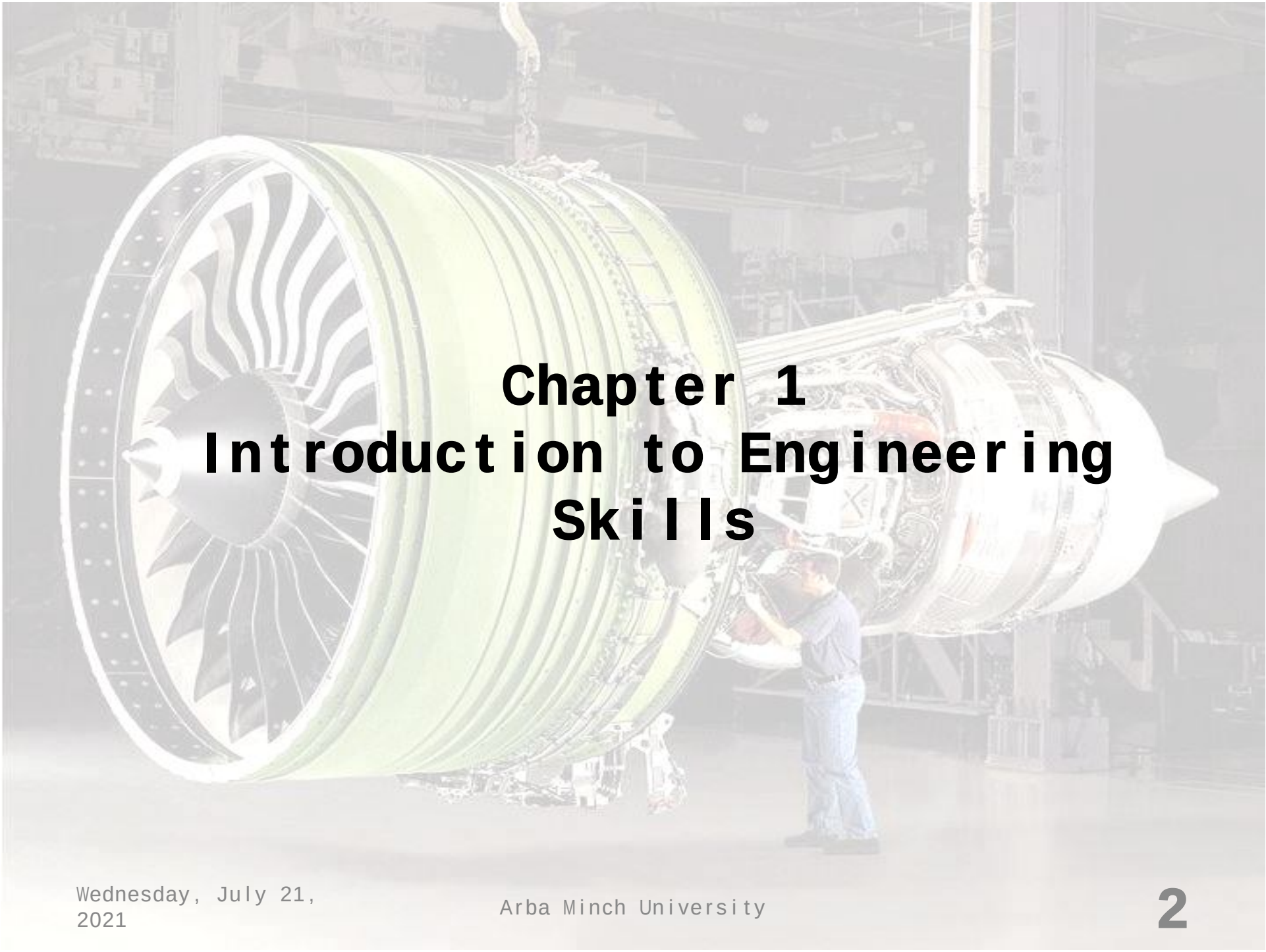
Arba Minch Institute of Technology
Faculty of Electrical and Computer Engineering

Introduction to Engineering Profession(GENG-2061)

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July.2021

Arba Minch University



Chapter 1
Introduction to Engineering Skills

1.1 Engineering Definition & Thinking

- **Engineering**: is the creative process of applying
  - Scientific and mathematical principles
  - Experience
  - Judgment /common sense
    - to address a need(problem)
    - that results in a **new product, process or system**
- Engineers use their knowledge of
  - science,
  - mathematics,
  - logic,
  - economics, and
  - appropriate experience

  to find suitable solutions to a problem.
- Creating **an appropriate mathematical model of a problem** allows them to
  - analyze it (sometimes definitively), and
  - to test potential solutions.

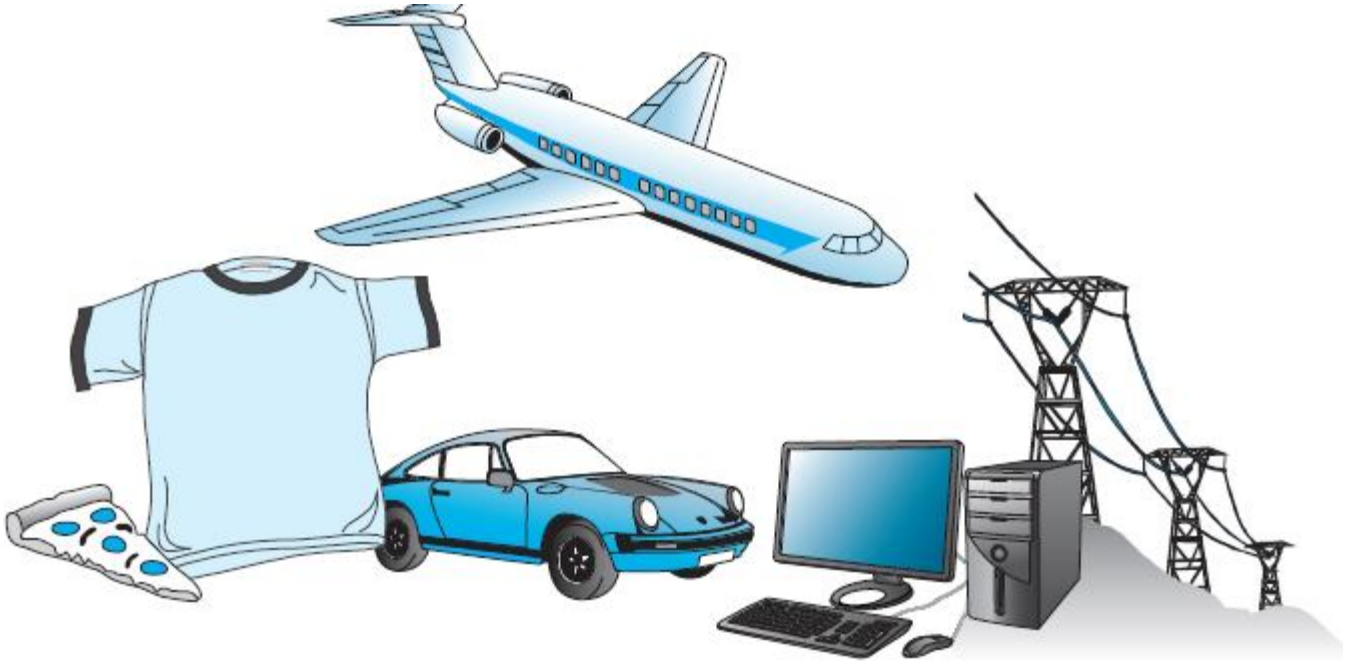
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- Usually **multiple reasonable solutions** exist, so engineers must
  - evaluate the different design choices on their merits and
  - choose the solution that **best** meets their requirements

- The discipline of engineering is extremely broad, and encompasses a range of more specialized fields, each with a more specific emphasis on
  - particular areas of technology and
  - types of application.

Continued…

- Engineers make **products** and provide **services** that make our lives better



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Engineer vs. Technologist

- **Engineer**
  - Develops conceptual design
  - Research and application of scientific concepts
  - Design of new products and systems

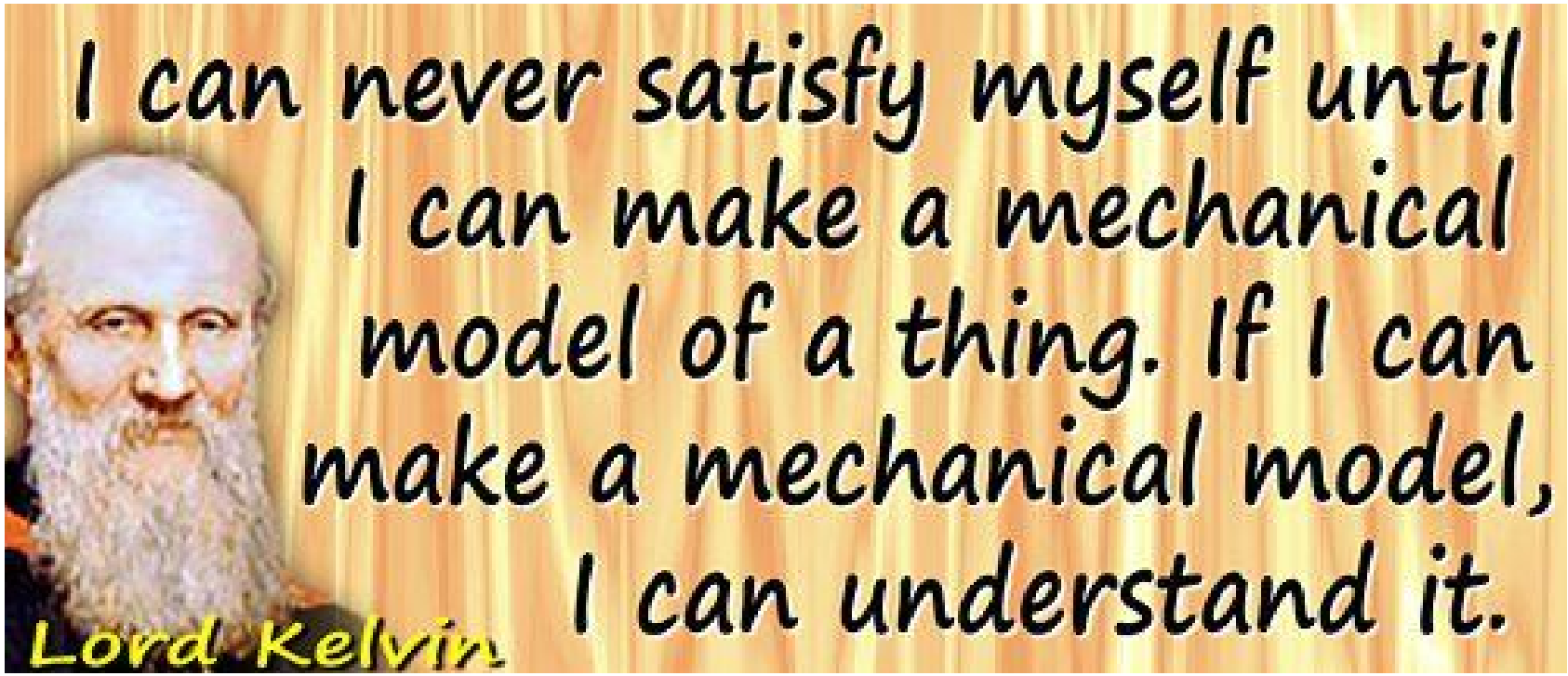
- **Technologist**
  - Shares **theoretical** background with engineer
  - **Practical** experience in application , implementation , sales of engineering design

1.2 Attributes of an engineer

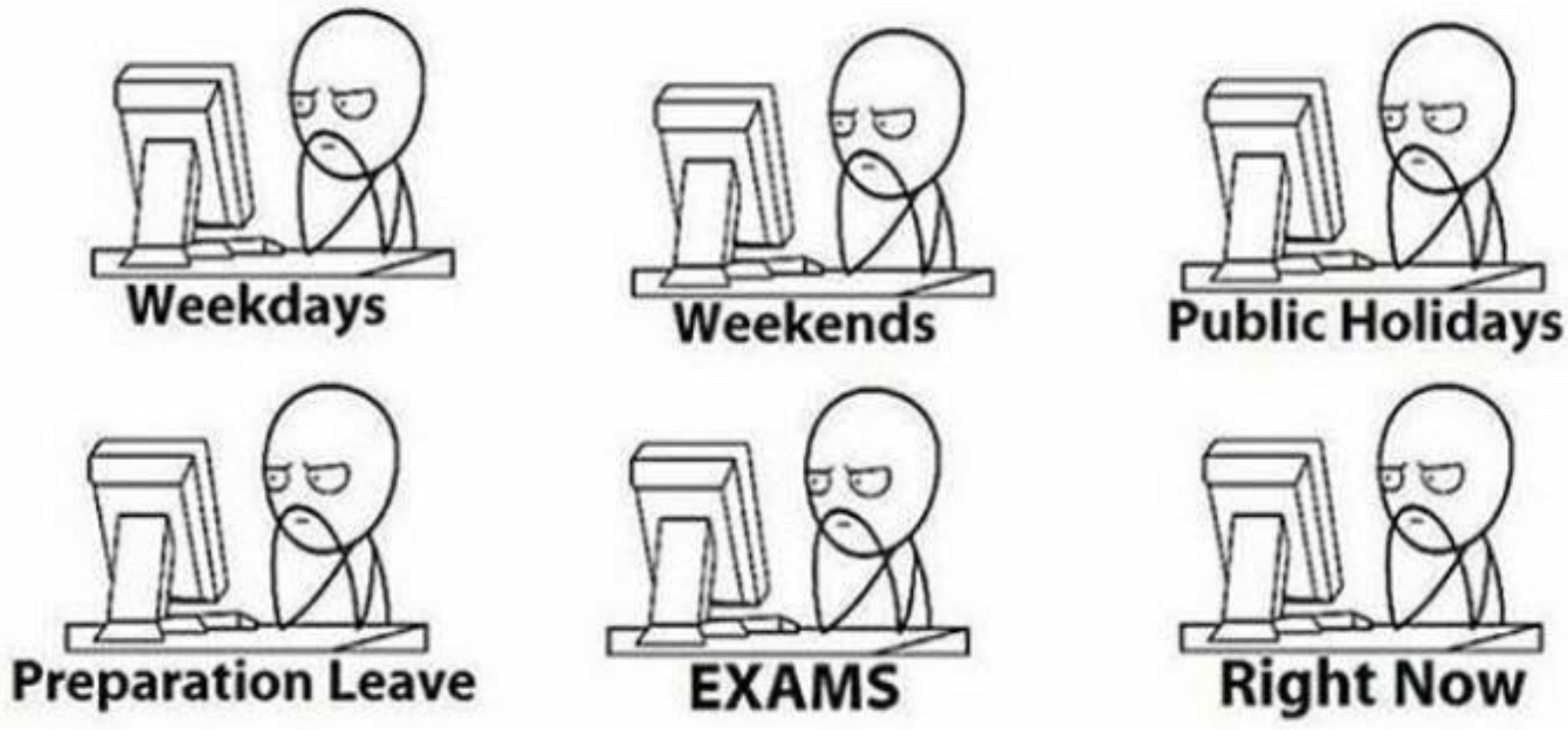
- ability to apply knowledge of **mathematics, science, and engineering;**
- ability to design and conduct **experiments**, as well as to analyze and interpret data
- ability to design a system, component, or process to meet **desired needs**;
- ability to function on **multidisciplinary** teams;
- ability to identify, formulate, and solve **engineering problems**;
- understanding of **professional and ethical responsibility**;
- ability to **communicate effectively**;

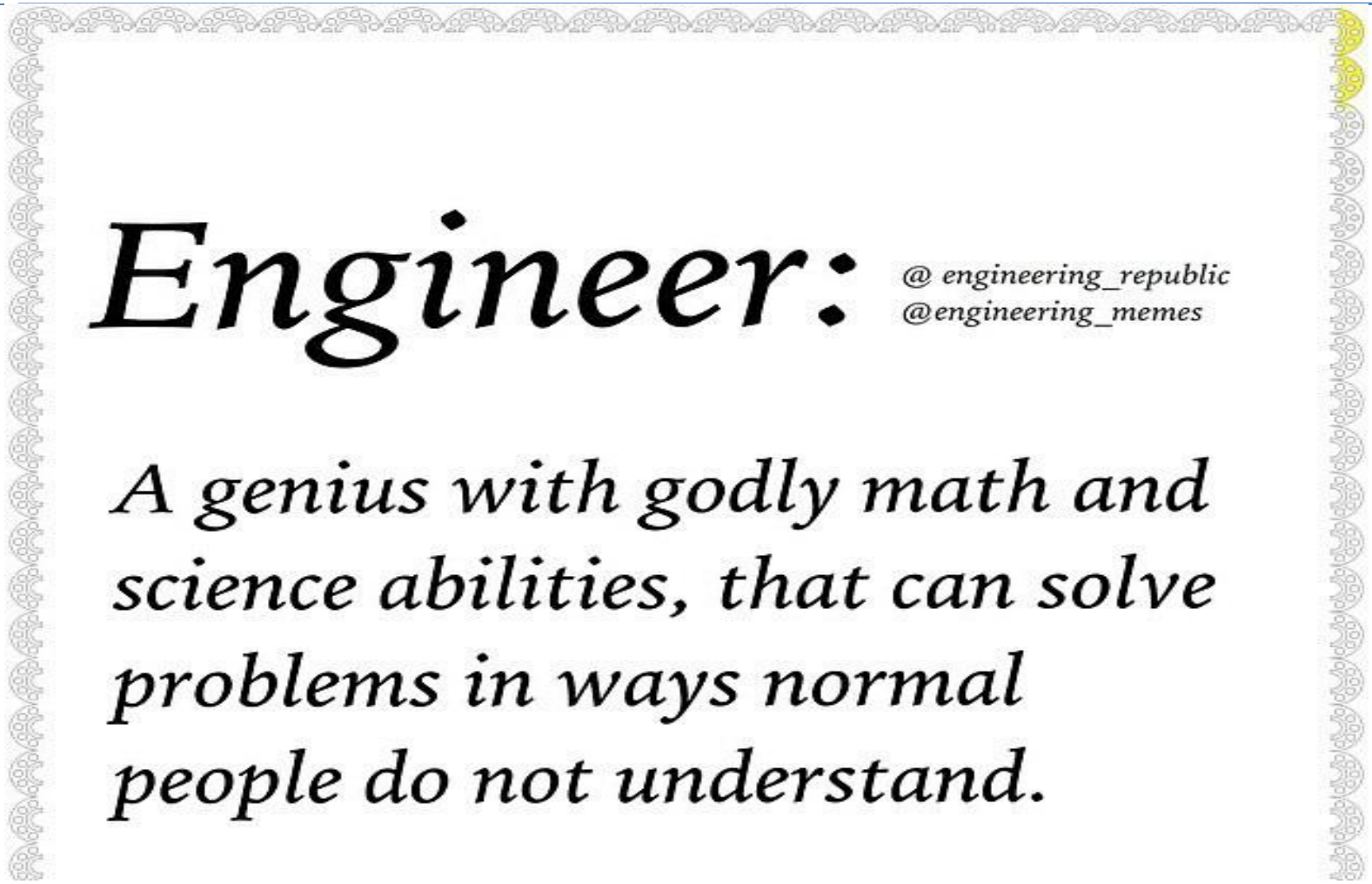
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- the **broad education** necessary to understand the impact of engineering solutions in a global and societal context;
- a recognition of the need for and an ability to engage in **lifelong learning**;
- a knowledge of **contemporary issues**;
- ability to use the **techniques, skills, and modern engineering tools** necessary for engineering practice.
- An ability to get **updated with the advancement of technology**


I can never satisfy myself until I can make a mechanical model of a thing. If I can make a mechanical model, I can understand it.
Lord Kelvin

ENGINEERS DURING :



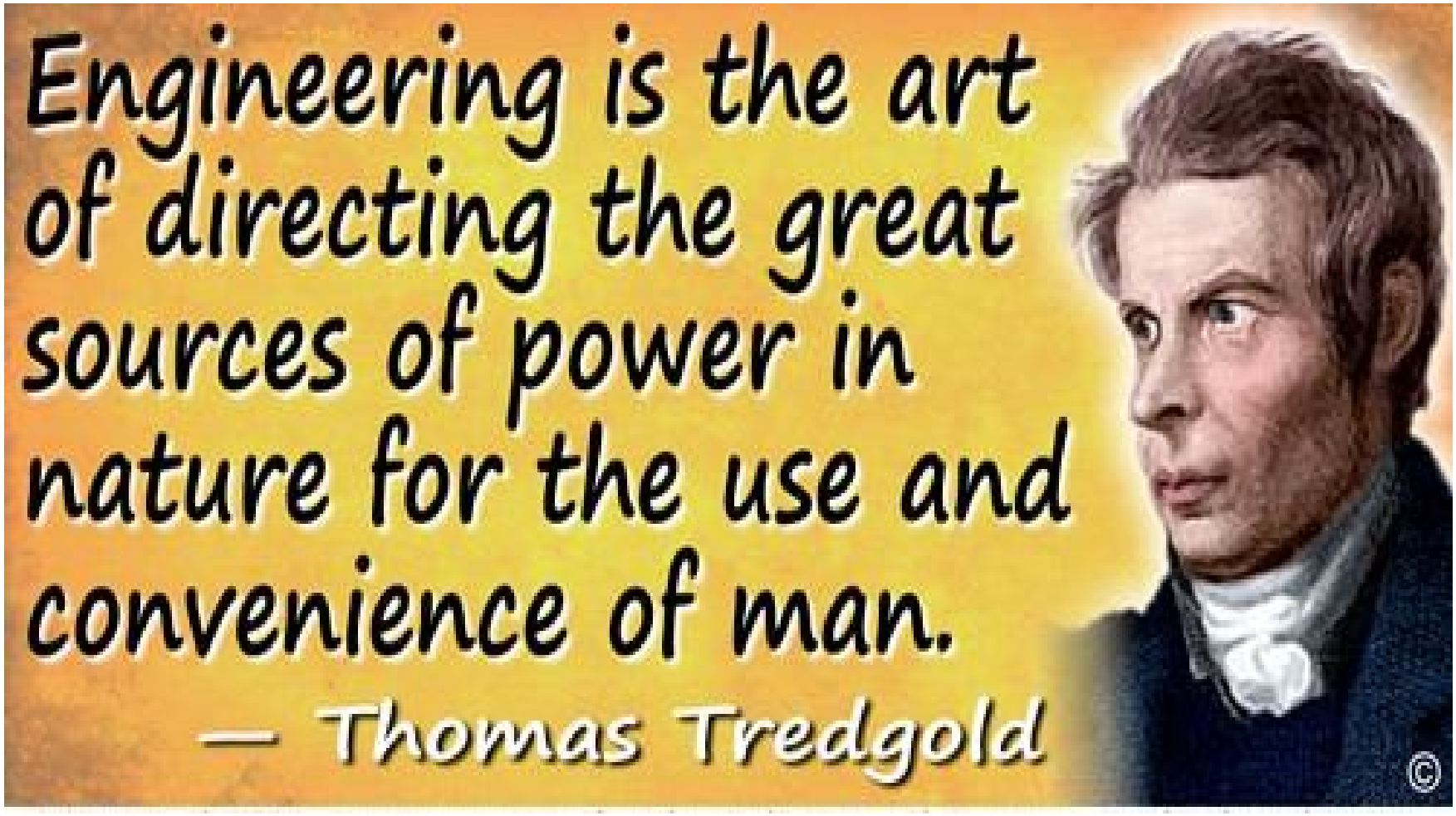

Engineer:
@ engineering_republic
@engineering_memes
A genius with godly math and science abilities, that can solve problems in ways normal people do not understand.

1.3 Resource and their utilization

- Engineers use different resources to simplify human's life.
- Energy as an engineering resource.
- The different form of energy resources can be utilized by the engineers and then can be converted into different forms which are suitable for the particular application.

Type of energy	Resources
Kinetic energy	wind
Gravitational energy	hydro(water falls)
Geo thermal energy	hot springs
Electrical energy	Electricity
Magnetic energy	electricity
Radiation energy	sun
heat energy	coal ,gas ,petroleum

Continued…


Engineering is the art of directing the great sources of power in nature for the use and convenience of man.
— Thomas Tredgold

1.3.1 Kinetic Energy-wind

- Wind energy is the kinetic energy of air in motion, also called wind.
- **Wind power** is the conversion of wind energy into a useful form of energy, such as using wind turbines to produce electrical power, windmills for mechanical power, wind pumps for water pumping or drainage, or sails to propel ships.
- Large wind farms consist of hundreds of individual wind turbines which are connected to the electric power transmission network.



1.3.2 Gravitational energy : hydro(water falls)

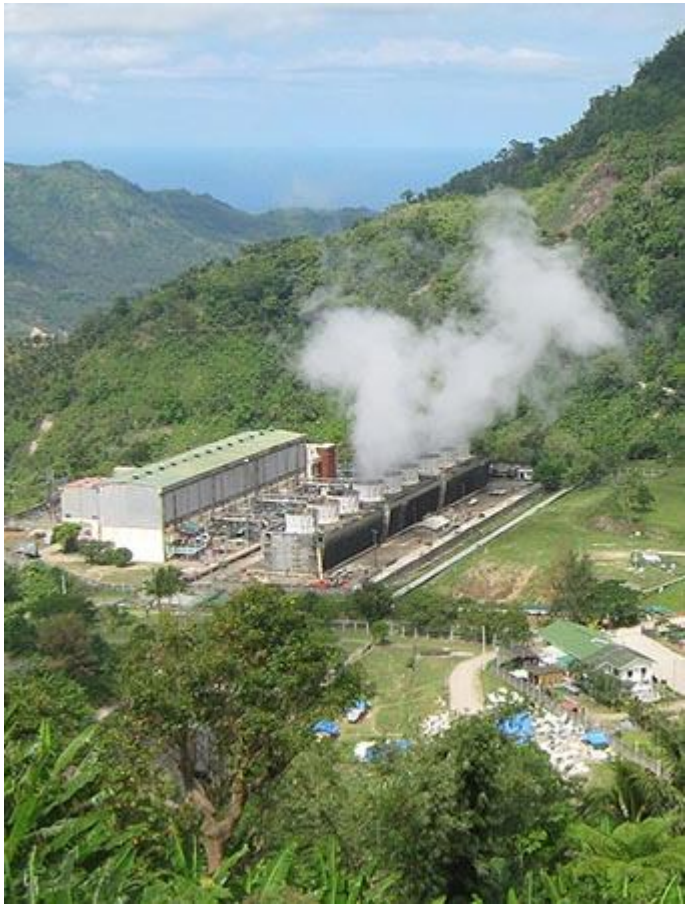
- Hydro-power or water power is power derived from the energy of falling water due to gravitational force, which may be harnessed for useful purposes.
- Since ancient times, hydro-power has been used for irrigation and the operation of various mechanical devices, such as watermills, sawmills, textile mills, domestic lifts, power houses



The Tekeze Arc Dam, one of the hydro plants completed in 2009, generates 300 MW.

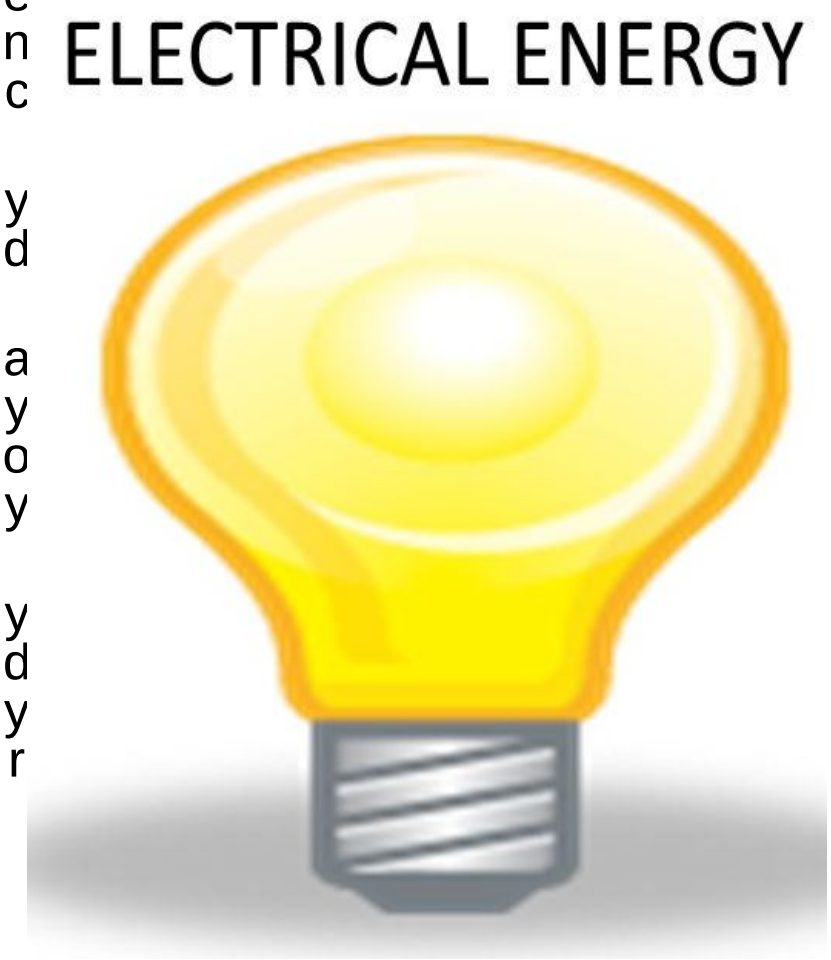
1.3.3 Geothermal energy : hot springs

- A **hot spring** is a spring that is produced by the emergence of geothermal heated groundwater from the Earth's crust.
- The water issuing from a hot spring is heated by geothermal heat, i.e., heat from the Earth's mantle.
- In general, the temperature of rocks within the earth increases with depth.
- If water percolates deeply enough into the crust, it will be heated as it comes into contact with hot rocks.
- The steam available from the springs can be utilized for power generation by the help of a steam turbine(creates mechanical energy from steam) and an alternator(ac generator converts mechanical energy to electrical energy).



1.3.4 Electrical energy: Electricity

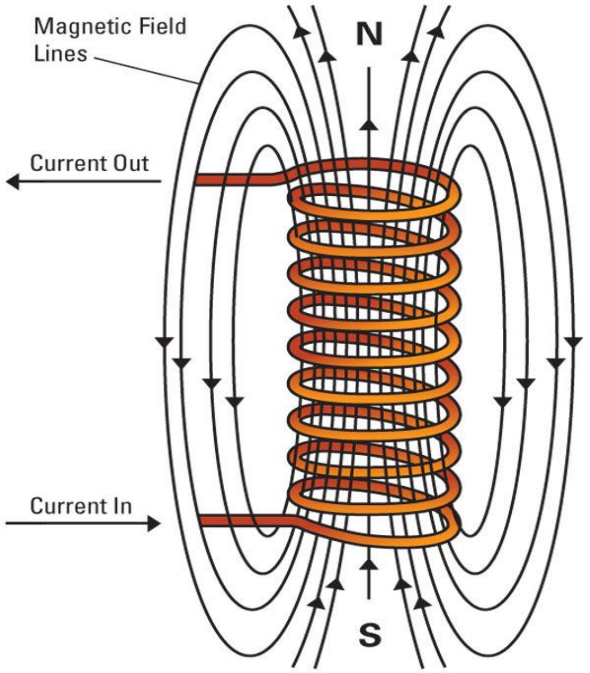
- The electricity is produced from the power station and is available at all the places like in homes, industries, public places etc.
- The electrical energy available can be converted into different forms
- For example if we use a fan the electrical energy is converted into mechanical and finally into wind energy.
- The electrical energy available can be converted into heat energy by electric stove, water heater, iron box etc.



1.3.5 Magnetic energy: electricity

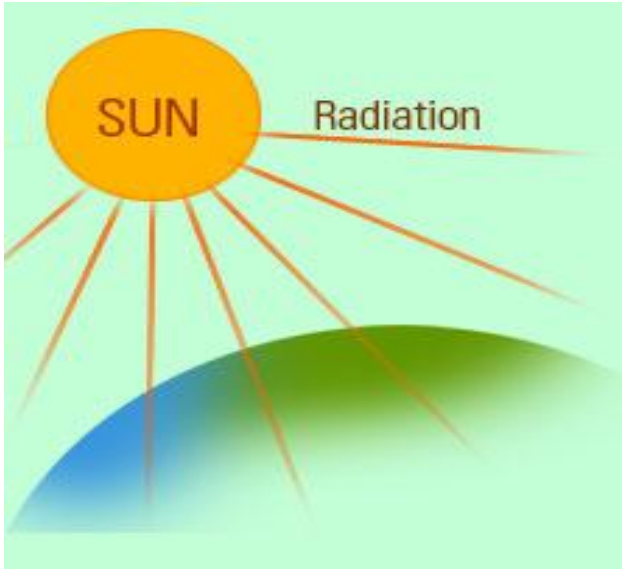
- The electricity can be used to create magnetic energy by connecting coils .
- the magnetic energy created can be used for several purposes.
- The working Principles of generators, transformers, motors are based on this conversion .





1.3.6 Radiation energy: sun

- The sun is source of unlimited energy and emits the energy in the form of radiation.
- The radiation energy can converted into heat energy(solar cookers),electrical energy(solar photo voltaic).



1.3.7 Heat energy : coal ,gas ,petroleum

- The heat energy can be created by burning of coal, petroleum and gas.
- The heat energy of coal is used in power plants for generation of steam which finally is converted into electrical energy.
- The heat energy from pertrol, gas is used for vechiles like bus, car, motor cycles to create mechanical energy.



1.4 Application of Engineering experience

- Historically ,engineering was largely empirical ,using experience to create new advances & technology
- Engineering allowed cultural and societal advances to occur
  - Water supply :Egyptian irrigation ,roman aqueducts, Dutch mills, Canals







Accomplishments of engineering

- Computers
- Radio
- Televisio
- Mobile
- Telephone
- Health technolo
- House hold app
- Aeroplane
- car
- Bus,
- train
- satellite













1.5 Engineering disciplines –classification

- Engineering, much like other science, is a broad discipline which is often broken down into several sub-disciplines.
- These disciplines concern themselves with differing areas of engineering work.
- Although initially an engineer will usually be trained in a specific discipline, throughout an engineer's career the engineer may become **multi-disciplined, having worked in several of the outlined areas**.
- Engineering fields are classified as follows
  - Electrical Engineering
  - Chemical engineering
  - Civil engineering
  - Mechanical engineering
  - Computer Engineering
  - Construction Technology& Management Engineering
  - Urban and Regional Planning Engineering
  - Genetic engineering
  - Food processing engineering,
  - Software Engineering
  - etc

Chapter 2
Electrical Engineering Disciplines

2.1 Introduction

2.1.1 Definition:-

- Electrical Engineering is branch of engineering that deals with the
  - ❖ design,
  - ❖ construction, and
  - ❖ practical uses of

  electrical systems with the study and application of
  - ❖ electricity,
  - ❖ electronics, and
  - ❖ electromagnetism.

2.1.2 Importance of Electrical Engineering

- The field first became an identifiable occupation in the late nineteenth century after commercialization of the
  - **electric telegraph** and
  - **electrical power supply.**

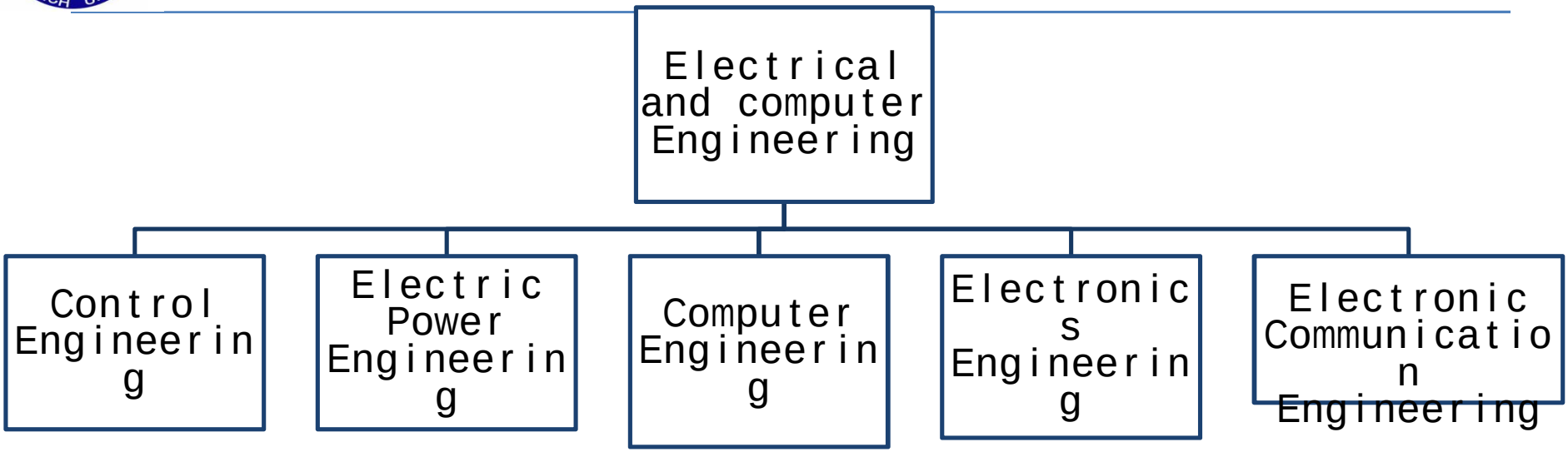
- Electrical engineering is a wide field of study which comprises all aspects of modern technological developments.

- At present, every aspect of day to day activities of people is highly tied with the field of electrical engineering.

Continued…

- Electrical engineers design and develop electrical systems and components focusing on
  - **economy,** **safety,**
  - **reliability,** **quality** and **sustainability**.
- They are involved in projects from the concept and detail of the design through to implementation, testing and handover.
- For example, they may work on the design of telecommunication systems, the operation of electric power systems, the lighting and wiring of buildings, the design of household appliances or the electrical control of industrial machinery.

2.2 Application areas & Disciplines



- Most of Electrical systems are made with the integration of those areas of electrical engineering.

2.2.1 Electric Power Engineering

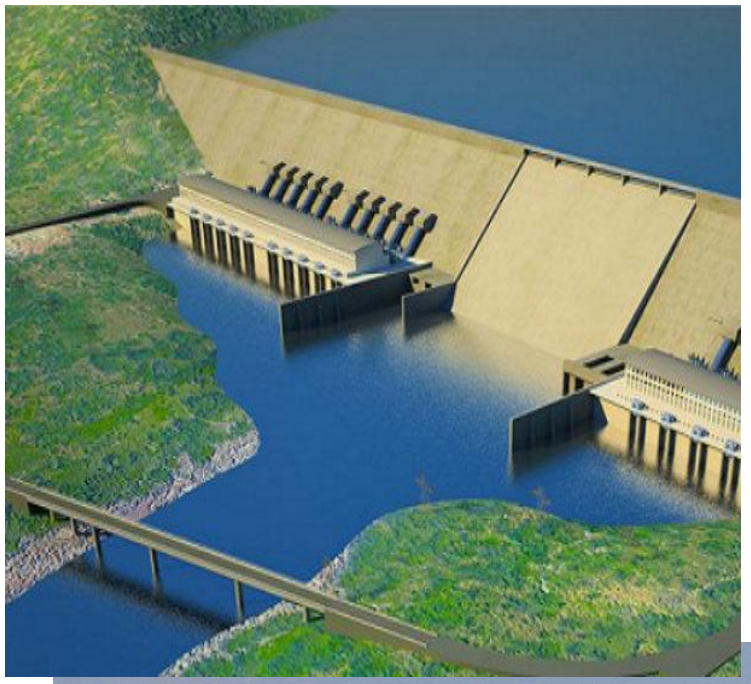
- Electric Power engineering deals with the generation, transmission and distribution of electricity as well as the design of a range of related devices including **transformers, electric generators, electric motors** and so on.
- In many regions of the world, governments maintain an electrical network called a power grid that connects a variety of generators together with users of the energy.
- Users purchase electrical energy from the grid, avoiding the costly exercise of having to generate their own.
- Electric Power Engineers may work on the **design** and **maintenance** of the power grid as well as the power systems that connect to it.
- Electric Power engineers may also work on systems that do not connect to the grid, called off-grid power systems.

Continued…

- Electric power is converted from different primary source of energy in to electrical energy
- In power systems, there are three main sectors, namely
  - Electric power generation engineering
  - Electric power transmission engineering
  - Electric power distribution engineering

Electric power generation

- it deals with generation of electricity.
- The place where power is generated is called generating station(power plant).
- Power production (Generation station), consists of the following main parts:
  - prime mover (creates mechanical energy, examples are steam turbines, hydel turbines)
  - generator (converts mechanical energy into electrical energy)
  - step-up transformer (in order to transmit power over longer distances the generated voltage is stepped up by this device to reduce power losses), circuit breaker(a big switch to connect or disconnect an electric circuit)





Electric power transmission engineering

- It deals with transmission of power from generation to distribution
- Transmission station, consists of the following main parts:
  - towers (gives mechanical support to conductors)
  - conductors(carry electric power)
  - insulators(insulate conductor and tower)
  - switching stations
    - circuit breakers
    - transformers





Electric power distribution engineering:

- it deals with the distribution of power from transmission system to the consumers or loads
- Distribution station, consists of the following main parts:
  - step-down transformers(the high voltage of transmission system is stepped down by this device for connecting it to distribution system)
  - Feeders(feeder is a conductor which feeds or supplies power to consumers or load)
  - circuit breakers.



Application areas

- Power grids
- Renewable energy sources
- power engineers work will include generators, transformers, circuit breakers, relays and transmission lines
- Substations
- Designing of electric motors

Jobs

- **Ethiopian electrical power corporation**
- **.power grid**
- substations

2.2.2 Control Engineering

- Control engineering focuses on the
  - **modeling of a diverse range of dynamic systems and**
  - **the design of controllers that will cause these systems to behave in the desired manner.**
- To implement such controllers, control engineers may use
  - electrical circuits,
  - digital signal processors, microcontrollers and PLCs (Programmable Logic Controllers).
- Control engineering has a wid range of applications in industrial automation systems.





Continued…

- There are two types of control systems based on devices used in them. They are
  - **Analog control system**: it is implemented by the use of hardware

Example: Electric stove with Knob control is analog control system

  - **Digital control system :** it is implemented by the use of computers, micro controllers, P.L.C s.

Application Areas

- Industrial automation systems
- Industrial areas include quality control of manufactured products
  *Automatic assembly line
  *Machine tool control
- Computer control
  *Power Systems
  *Robotics
  *Nano technology

Application areas

- Temperature control systems
  *Steering control of automobiles
- Voltage stabilizer systems
  *Idle speed control of automobiles
  *Industrial sewing machine

Jobs

- Industrial area includes:-
- Textile
- Sugar
- Bewerages

2.2.3 Instrumentation Engineering

- Instrumentation engineering deals with the design of devices to measure physical quantities such as pressure, flow and temperature, weight, etc.
- The design of such instrumentation requires a good understanding of physics that often extends beyond electromagnetic theory.
- Instrumentation for industries is responsible for manufacturing the devices which measure the parameters directly or indirectly
  - Devices that make direct measurements: temperature ,pressure etc
  - Devices that make indirect measurements: Strain guage (indirect measurement of weight) manometer (indirect measurement of pressure)
  - Instrumentation for medical :This include
    - X-ray
    - Ultra sound devices
    - Magnetic resonance imaging(MRI)







Application Areas

- Medical area includes
- X-Ray
- ultra sound devices
- Magnetic Resonance imaging
- . Aerospace Industries: – For Dynamics and Control.
- 2 . Biomedical Industries.
- 3 . Nuclear Industries.
- 4 . Oil and gas industries.

Jobs

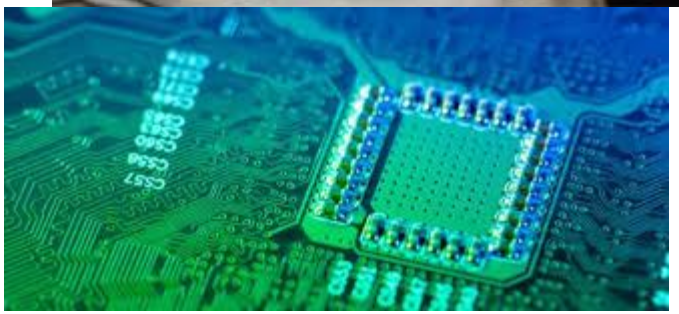
- Industries
- Hospitals

2.2.4 Computer Engineering

- Computer engineering deals with the design of computers and computer systems.
- This may involve the design of new hardware or the use of computers to control an industrial plant.
- Computer engineers may also work on a system's software.
- However, the design of complex software systems is often the domain of software engineering, which is usually considered as a separate discipline.







Application areas

- Computers and computing systems applicable in nearly every aspect of modern life, from automobile systems to air traffic control systems, missile guidance systems, surgical equipment, and portable devices
- Artificial Intelligence
- Robotics
- Cloud computing

Application areas

- Image procesing
- Speech processing
- Data mining
- Cyber security
- Wireless networking
- Design automation
- Mobile computing
- Embedded systems

Job opportunities

- software Engineer
- Web designer
- Web developer
- Hardware Engineer
- Multimedia programmer
- Database administrator
- IT consultant
- Data analysist
- Database administrator
- Network manager
- Cyber security analyst

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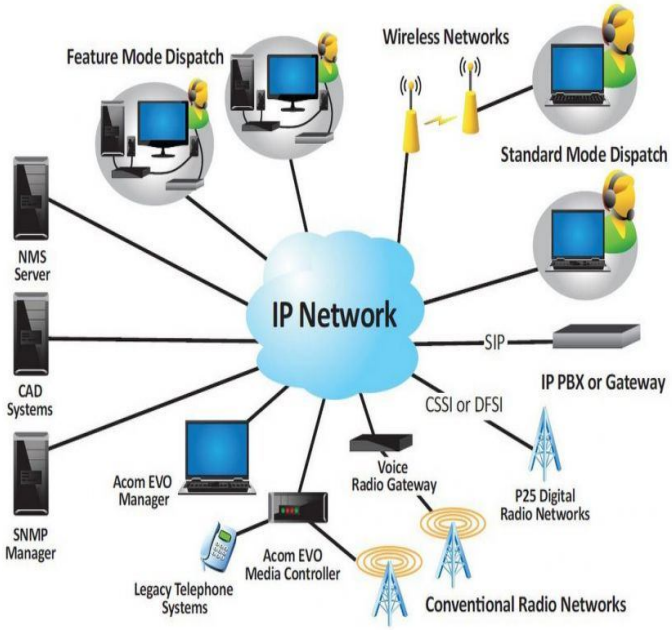
- **Software engineering-**the application of a systematic approach to the **development, operation, and maintenance of software** or the application of engineering to software. Examples windows 8 in desktop computer, PC games
- **Hardware engineering-**designing, developing, and testing various computer equipment. It can range from circuit boards and microprocessors to routers.
- **Network engineering-**designing and maintaining computer networks, such as corporate networks or the Internet. Examples are LAN (local area network),WiFi (wireless LAN)

2.2.5 Electronic engineering

- Electronic engineering deals with the design and testing of electronic circuits that use components such as resistors, capacitors, inductors, diodes and transistors to achieve a particular functionality.
- The tuned circuit, which allows the user of a radio to filter out all but a single station, is just one example of such a circuit.

2.2.6 Communication Engineering

- Communication engineering focuses on the transmission of information across a channel such as a coaxial cable, optical fiber or free space.
- In modern times, communication engineering typically involves the use of electronic **transceivers** such as the telephone, television, radio or computer.
- A typical communication system consists of three basic elements:
  - **Transmitte**r- that takes information and converts it to a signal
  - **Transmission medium-**that carries the signal
  - **Receiver-** that receives the signal and convert it into information





Communication Engineering

- Typical examples are high-speed networks, Internet, cellular and satellite ommunications, and WiFi or wireless area networks.
- Telecommunication Engineering concentrates on optical signal processing and networks, mobile communications, microwave techniques and radiation from ICs and PCBs

Application areas

- Wireless telephony services
- - Radio and satellite communications
- - Internet and broadband technologies
- Global position system

Jobs

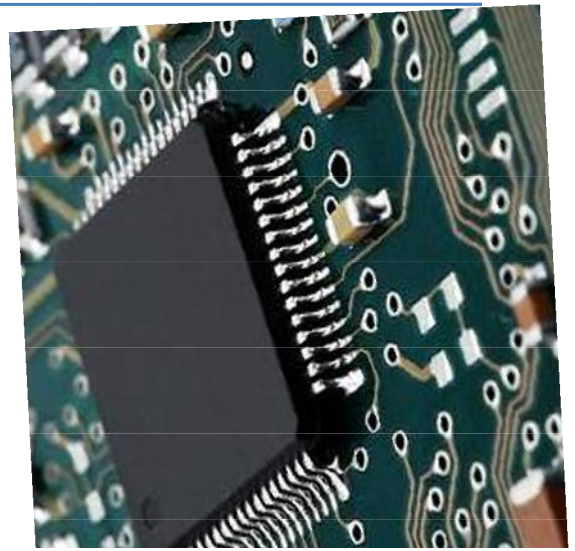
- Cell phone companies
- Ethiopian telecommunication
- Radio and TV broadcasting
- Airports
- Radars
- Wireless networking side

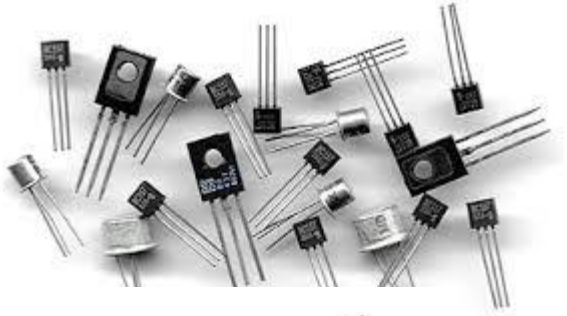
2.2.7 Microelectronic Engineering

- Microelectronic engineering deals with the design and fabrication of **very small electronic circuit components** for use in an integrated circuit or sometimes for use on their own as a general electronic component.

- The most common microelectronic components are semiconductor transistors, although all main electronic components (resistors, capacitors, inductors) can be created at a microscopic level.

- The field of microelectronics involves a significant amount of chemistry and material science and requires the electronic engineer working in the field to have a very good working knowledge of the effects of quantum mechanics.







Application Areas

- Manufacturing of semiconductor chips
- Integrated circuits
- Mobile manufacturing industries

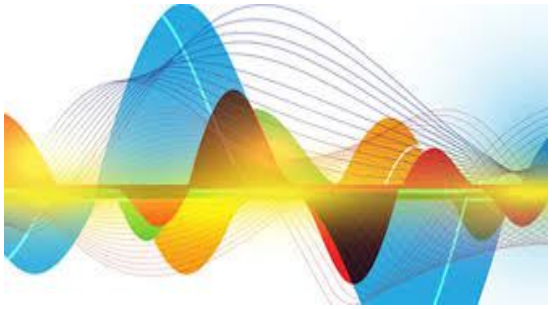
Jobs

- Mobile companies
- Computer manufacturing industries
- All electronic manufacturing industries

2.2.8 Signal Processing Engineering

- Signal processing deals with the analysis and manipulations of signals.
- Signals can be either analog, in which case the signal varies continuously according to the information, or digital, in which case the signal varies according to a series of discrete values representing the information.
- For analog signals, signal processing may involve the amplification and filtering of audio signals for audio equipment or the modulation and demodulation of signals for telecommunications.
- For digital signals, signal processing may involve the compression, error detection and error correction of digitally sampled signals.



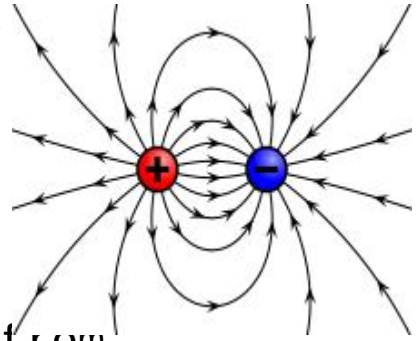


END OF CHAPTER TWO!

Chapter 3
Introduction to Electrical Engineering

3.1 Electricity

- **Electricity**: **Electricity** is the set of physical phenomena associated with the presence and flow of electric charge.
- **Charge**: is the physical property of matter that causes it to experience a force when close other electrically charged matter.
- There are two types of electric charges
  - positive and
  - negative.



- Positively charged substances are repelled from other positively charged substances, but attracted to negatively charged substances; negatively charged substances are repelled from negative and attracted to positive.
- Charge is measured in coloumbs (c)

  1 electron charge = 1.602×10^{-19}C,

  1 C= $1/(1.602\times10^{-19})=6.24\times10^{18}$ electrons

3.1.1 Current

- **Current**: current is the rate of flow of charge, measured in amperes(A) and generally indicated by letter i

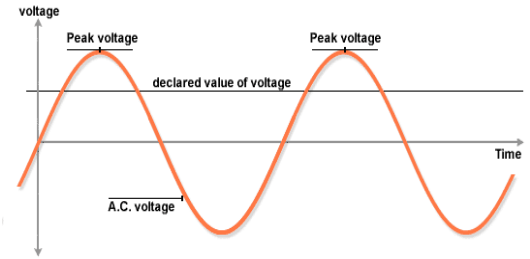
$$i = \frac{\Delta q}{\Delta t} = dq/dt$$

$$1A = 1c/1sec$$



Direct Current Graph

- Two types of currents: direct and alternating
- A direct current (dc) is a current that remains constant with time.
- An alternating current (ac) is a current that varies sinusoidally with time.



3.1.2 Voltage:

- To move the electron in a conductor in a particular direction requires some work or energy transfer.
- This work is performed by an external **electromotive force** (emf).This emf is also known as *voltage* or *potential difference*.
- The voltage V_{ab} between two points *a* and *b* in an electric circuit is the energy (or work) needed to move a unit charge from *a* to *b*; mathematically,

$$V_{ab} = dw/dq$$

- Voltage (or potential difference) is the energy required to move a unit charge through an element, measured in volts (V).

  1 volt = 1 joule/coulomb = 1 newton meter/coulomb

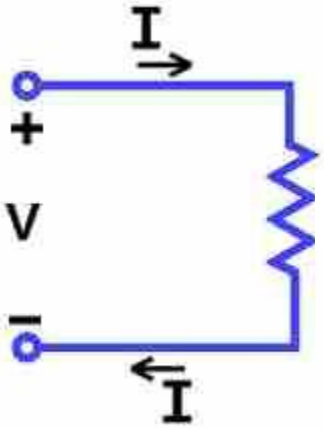
3.1.3 Power

- Power is the time rate of delivering or absorbing energy, measured in watts (W).

$$P=dw/dt$$

where P is power in watts (W), w is energy in joules (J), and t is time in seconds (s).

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = VI$$



- The power absorbed or supplied by an element is the product of the voltage across the element and the current through it.

3.1.4 Energy

- Energy is the capacity to do work, measured in joules (J).

$E = \int p dt$ for time varying power and

$E = pt$ for time constant power

- The electric power utility companies measure energy in watt-hours (Wh) or in kilo watt-hours(kWh),where

1 Wh = 3,600 J and 1kWh=1000Wh

3.2 Basic Laws of Circuit Analysis

- Ohm's Law
- Kirchoff's Law

3.2.1 Ohm's Law

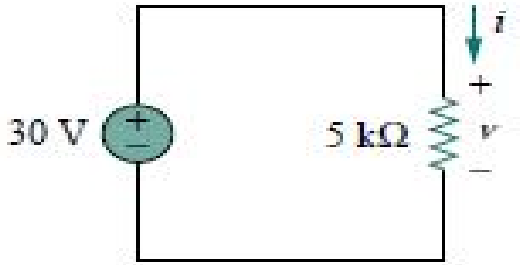
- **Ohm's Law** states that the voltage *v across a resistor is directly proportional* to the current *i flowing through the resistor.*
- That is, $v \propto I$
- Ohm defined the constant of proportionality for a resistor to be the resistance,*R. (The resistance is a material property which can change if the* internal or external conditions of the element are altered, e.g., if there are changes in the temperature.) Thus, Eq. above becomes , $v = iR$

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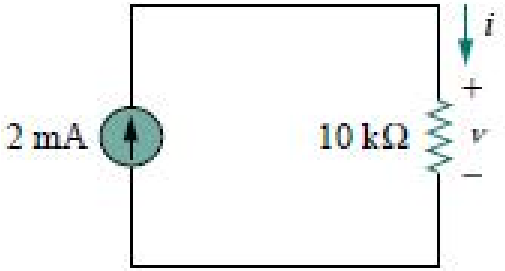
- which is the mathematical form of Ohm's law. *R in Eq. above is measured* in the unit of ohms, designated Ω. Thus,
- The resistance *R of an element denotes its ability to resist the flow* of electric current; it is measured in ohms (Ω).
  Example:-
- An electric iron draws 2 A at 120 V. Find its resistance.
- The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy. How much current is drawn by a toaster with resistance 12Ω *at 110 V?*

Continued…Example

- In the circuit shown in Fig.below ,calculate the current *i, the conductance G, and the power p.*



- For the circuit shown in Fig. below, calculate the voltage *v, the conductance G, and the power p.*



3.2.2 Kirchoff's Law

- Ohm's law by itself is not sufficient to analyze circuits. However, when it is coupled with Kirchhoff's two laws, we have a sufficient, powerful set of tools for analyzing a large variety of electric circuits. Kirchhoff's laws were first introduced in 1847 by the German physicist Gustav Robert Kirchhoff (1824–1887). These laws are formally known as Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL).
- Kirchhoff's first law is based on the law of conservation of charge, which requires that the algebraic sum of charges within a system cannot change.

Continued…

- Kirchhoff's current law (KCL) states that the algebraic sum of currents entering leaving a node (or a closed boundary) is zero.
- Mathematically, KCL implies that

$$\sum_{n=1}^{N} i_n = 0$$

- where *N is the number of branches connected to the node and in is the nth current entering (or leaving) the node. By this law, currents entering* a node may be regarded as positive, while currents leaving the node may be taken as negative or vice versa.

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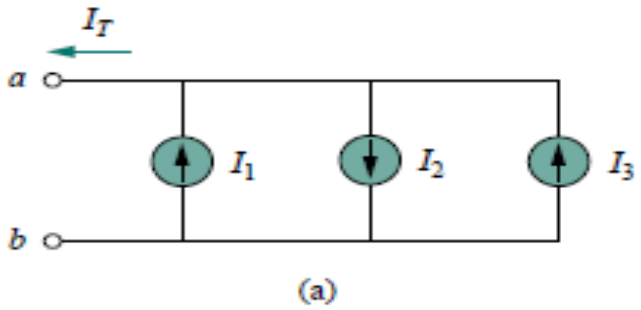
- The sum of the currents entering a node is equal to the sum of the currents leaving the node.
- Kirchhoff's second law is based on the principle of conservation of energy:
- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path(or loop) is zero.
- Expressed mathematically, KVL states that

$$\sum_{m=1}^{M} v_m = 0$$

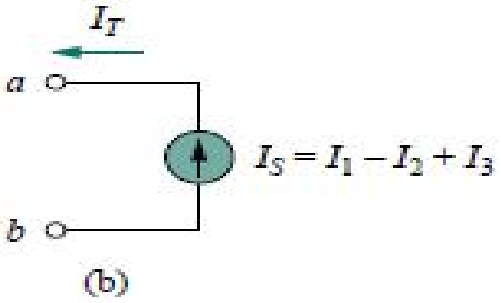
- where*M is the number of voltages in the loop (or the number of branches* in the loop) and *vm is the mth voltage.*

Continued…

- For example, the current sources shown in Fig.below(a) can be combined as in Fig.below(b). The combined or equivalent current source can be found by applying KCL to node *a*.
- *IT + I2 = I1 + I3* or
- *IT = I1 – I2 + I3*



(a)



(b)

Continued…

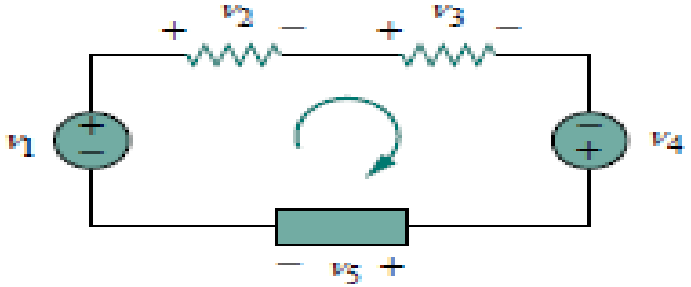
- For example, as we reach branch 3, the positive terminal is met first; hence we have+*v3. For branch* 4, we reach the negative terminal first; hence, *-v4. Thus, KVL yields*

$$-v1 + v2 + v3 - v4 + v5 = 0$$

- Rearranging terms gives

$v2 + v3 + v5 = v1 + v4$ which may be interpreted as

Sum of voltage drops = Sum of voltage rises



END OF CHAPTER THREE!